

Dharma Events

Feature & Build Summary

A complete overview of what has been built: core platform, plus every post-launch enhancement.

This document summarizes the Dharma Events registration and QR check-in platform for a non-technical audience. For full technical build logs see `docs/IMPLEMENTATION_STATUS.md` in the source code; for step-by-step usage instructions see the companion **User Testing Guide**.

Core Platform

A self-hosted, browser-based (installable as a mobile app / PWA) system for managing event registrations, emailing QR-coded invitations, and checking guests in at the venue — deployed via Docker on your Synology NAS.

Core capabilities

- Role-based accounts: Admin, Event Manager, Supervisor, Volunteer.
- Event lifecycle: Draft → Active → Completed → Archived, with independent Registration-open / Check-in-open toggles.
- Categories per event (e.g. General, VIP, Volunteer) to group registrations.
- Manual registration entry, plus bulk import from Excel/CSV with row-by-row validation (Valid / Warning / Error).
- Automatic QR-coded PDF ticket generation and email delivery, with retry-on-failure.
- Camera-based QR scanning for check-in, with partial/group check-in support and a manual search fallback.
- Live dashboard: registered/arrived/remaining counts, by-category and by-volunteer breakdowns, arrival timeline.
- Downloadable CSV/XLSX reports: attendance, no-shows, check-in transactions, invitation delivery.
- Audit logging of key admin actions.
- Production-hardened for the Synology NAS deployment: internal-only database network, structured logging, backup/restore scripts, and load-tested at 1,000+ registrations per event.

Post-Launch Enhancements

After going live, the following improvements were added in response to real usage and feedback:

Registration

- Mobile number field with country code.
- Optional "linked volunteer" per registration (e.g. the volunteer who invited/escorted a guest).

Invitations

- Attach a PDF (invitation letter, brochure, map) to an event; it's merged into the emailed ticket.
- Per-category PDF templates — a category can have its own attachment, overriding the event's common one.
- Linked volunteers can be automatically CC'd on their guest's invitation email (admin-configurable).
- One QR-coded ticket page per registered guest (e.g. 4 pax → 4 individually scannable pages), instead of a single shared page for the whole party.
- Admin "Preview PDF" shows exactly what will be emailed, including the correct attachment.

Administration

- Volunteers can be permanently deleted, in addition to the existing soft-deactivate option.
- A downloadable, pre-filled Excel template for bulk registration import.
- "Back to Event" navigation links on Registrations, Invitations, Scanner, and Dashboard pages.

Bug Fixes

- Fixed the installed app (PWA) intercepting Reports CSV/XLSX links and PDF View/Preview links and redirecting to the home page instead of opening the file.
- Fixed the QR scanner failing to record a check-in on scan.
- Fixed a Docker/Node.js version mismatch that broke database migrations during container builds.

Known Limitation

A registration email that is well-formed but points to a non-existent or since-closed mailbox may still show as "Sent" if the receiving mail server accepts it before bouncing later — only immediate rejections (e.g. malformed addresses) are currently caught and marked "Failed". Deferred at the customer's request; would require bounce-mailbox monitoring or a pre-send address-verification service to close this gap.

Requested, Not Yet Built

- Scheduled/timed sending of invitations.
- Grouping and batch-sending invitations by category (e.g. send VIP now, General tomorrow).